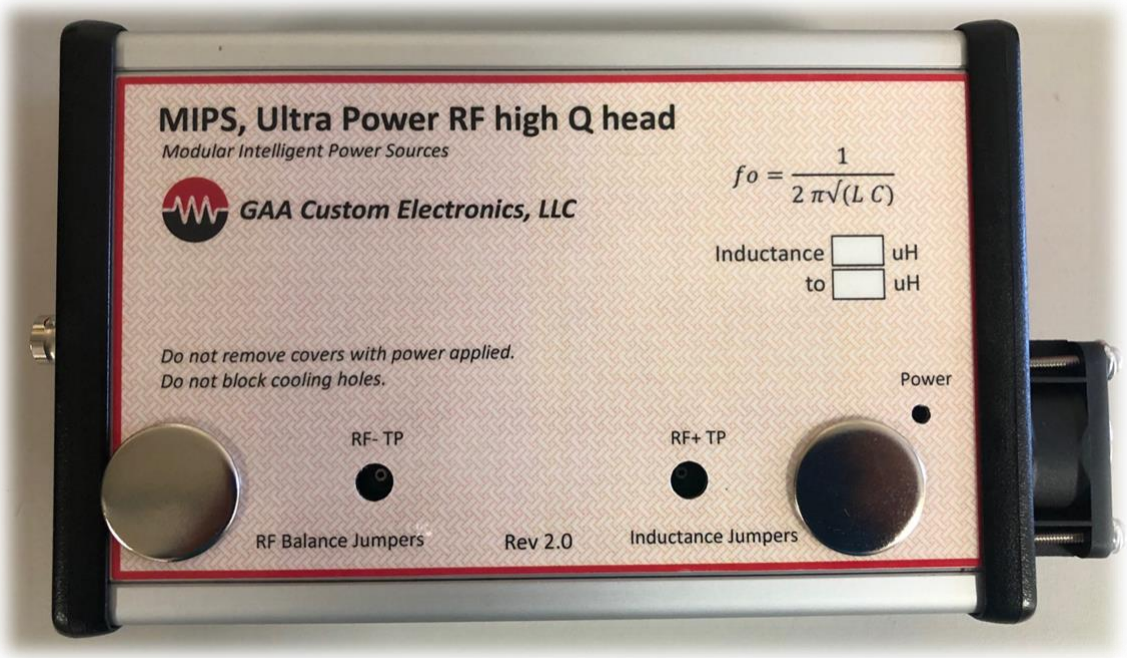
Ultra power RF head

Operations Manual
Rev 3.0, July 2, 2023



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WARNING
TO REDUCE THE RISK OF FIRE OR ELECTRICAL SHOCK DO NOT EXPOSE THIS EQUIPMENT TO RAIN OR MOSITURE.

WARNING
Only qualified electrical workers should operate and install this system.
Avoid spilling liquids onto/into the unit.
Do not expose to excessive heat or moisture.
Do not open – there are no user serviceable parts inside.
Do not block the chassis vent slots or the fan inlet.
Do not operate in an explosive environment.

Introduction

Thank you for purchasing High power RF head. This document describes the installation and operation of your High power RF head. The RF head is designed to interface with the MIPS system and is also compatible with PNNLs AMPS systems. This RF head is designed to drive ion transport devices such as ion funnels, quadrapoles, and even SLIM devices. This RF head has the ability to drive higher capacitance loads requiring higher power dissipation. With this RF head you are safe to operate up to 20 watts of power dissipation assuming the system has been tuned to the load. The MIPS system monitors this power and displays it on the user interface.

This document describes the operation and connection of this RF head to your system. This document also contains schematics and technical information.

During operation of the High power RF head it is important to leave open area around the air inlets and outlets to insure good air flow for cooling. This RF head is designed to operate in a laboratory environment. If your application has specific needs in terms of load capacitance and drive voltages and frequencies please contact us so we can design the RF head for your specific application.

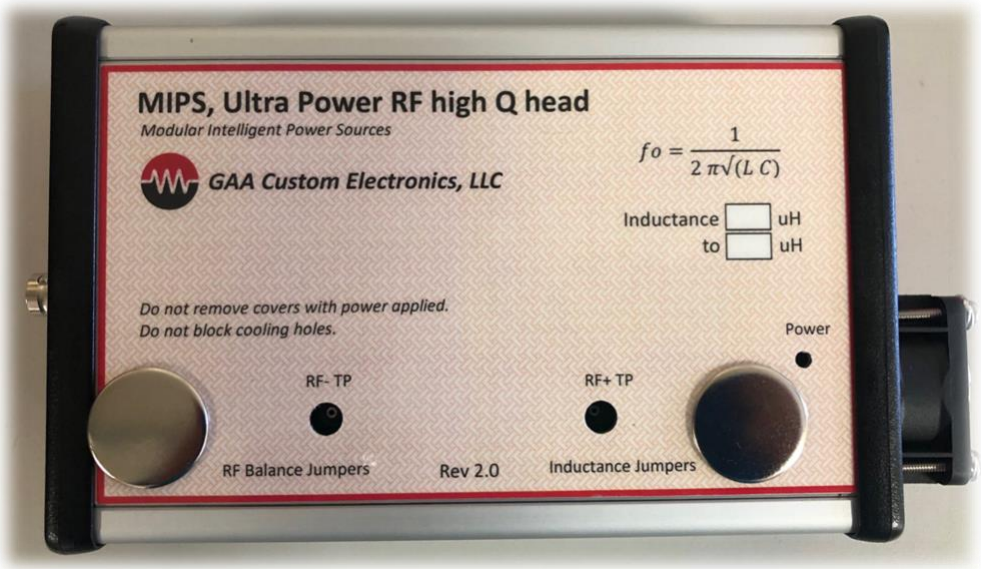
Options are available for external BNC inputs for DC bias as well as gating the RF on and off. Separate DC bias can be placed on each RF phase or a single DC bias for both phases. Gating can be performed with active high or active low TTL inputs. Please contact us to make sure we set the RF head up the way you need it for your application.

Connection and tuning of High Q RF heads

The RF high Q head generates a balanced output, two output phases with 180 degrees of shift between each phase. These phases connect to electrodes in ion guides. These electrodes present a capacitive load to the high Q head. The high Q head contains inductance that resonates with the ion guides capacitance. This resonance allows the high Q head to generate a few hundred volts peak to peak (Vpp).

The goal of this section is to provide general guidance to connect the RF high Q head to your system and a procedure for tuning your system.

Two BNC connectors on the RF high Q head provide the two phases of RF output and should be connected to your ion guide. The typical capacitance seen is these systems vary between 100 to 300 pF. The high Q head will resonate between 700 KHz to 3 MHz depending on the inductance of the head and your load capacitance.



RF high Q head. RF-TP and RF+TP are test points where you can insert a scope probe to monitor the RF signal. The BNC connectors are the RF output phases that should be connected to your RF device and optional bias and/or gate inputs.

The following procedure provides a step-by-step process that can be used to setup a new RF high Q head:

1.) With the MIPS power off connect the MIPS box to the high Q head using the 15 pin D cable provided with your system.
2.) The RF high Q head is connected with two BNC cables that you provide. Check the capacitance from the center of one cable to the center of the other then use the inductance of the high Q head and the formula on the high Q head to calculate the resonate frequency. This will give you a starting point for tuning. You do not have to measure the capacitance but if you can it will test the load and provide a starting point. Use a multi-meter and make sure the load is not shorted to ground or very low resistance. When you are sure the load is correct then connect the cables.
3.) Turn on the MIPS box and select the RF module. Set the drive level to 10 %.
4.) Adjust the frequency to achieve a maximum RF voltage. You can monitor the RF voltage using the reading on the MIPS box or use a scope and monitor the test points on the RF high Q head. Using a scope will also allow you to inspect the quality of the RF signals. I would recommend using a scope on initial setup but it is not necessary for routine operation. It is best to start tuning from a low frequency working toward higher frequencies. Monitor the RF head power and ensure it remains under a watt through this process. Depending on your load you could find multiple resonance points, always look for the strongest or highest peak voltage.
5.) After the peak is found you can increase the drive level as needed to reach the desired RF voltages. The resonate frequency will change with power level

so you will need to make fine adjustments in frequency as you increase drive. The resonate frequency may also change as your system warms due to capacitance changes etc.

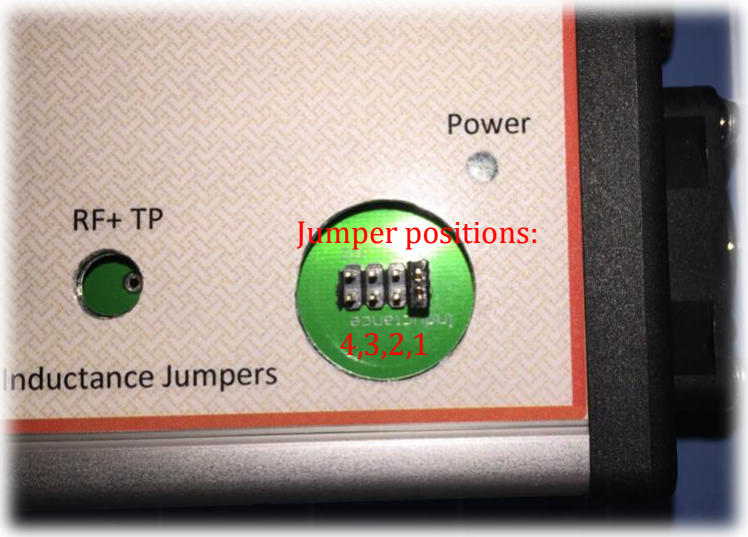
6.) Remove the scope and use the MIPS voltage readings to do the final tuning. The next step is to balance the two phases. If the RF voltage +Vpp and –Vpp do not match with in your requirements you can use the balance jumpers to make corrections. The scope needs to be removed because the capacitance of the probes can change the balance and can slightly change the resonate frequency. The front of the RF high Q head has hole plug covering a set of balance jumpers. These jumpers enable capacitance to be added to the –Vpp side of the RF head to ground. The figure below shows the jumpers and lists the capacitance options depending on the number of jumpers installed. Increasing the capacitance will lower the voltage on the –Vpp phase and raise it on the +Vpp phase. Changing the capacitance will also slightly change the resonate frequency. Adjust the jumpers to achieve the desired balance. Make sure the power is off before the jumpers are changed. Also make sure you save the RF module parameters before you turn off the power.

Install a jumper to add capacitance from the negative RF phase to ground. The capacitance for each jumper position is listed below. Multiple jumpers can be installed to increase the total capacitance.

1 = 12 pF
2 = 22 pF
3 = 47 pF
4 = 82 pF

Note! Install and remove jumpers with MIPS power off.

7.) The RF high Q heads inductance can be adjusted with in a limited range if the resonate frequency needs to change. There is a hole plug covering a jumper block on the front of the RF head as shown in the figure below. Its critical that only one jumper is installed on the jumper block. There are 4 inductance options on each RF head. Lower inductance results in higher resonate frequency. The RF heads highest efficiency is achieved at its highest inductance so you should try to operate at this inductance if possible.



Inductance selection jumpers. One and only one jumper must be installed. Highest output voltage is achieved with the jumper in position 1. Highest inductance is position 1, lowest is position 4.

Note! Install and remove jumpers with MIPS power off.

8.) After you have made all the needed adjustment and tuned you system allow it to operate for several minutes to stabilize. I recommend you then sent the drive level limit and power limit slightly above the operating levels of the system. This will provide a level of system protection.

This procedure provides general guidelines for tuning a High power RF high Q head and should help you perform general system setup. RF Ultra Q head power dissipation should not exceed 50 watts. In most applications you will only need 5 to 10 watts of power to produce the needed voltages.

The MIPS system supports automatic tuning of the RF head, this is a very simple procedure, the system will automatically set the best operating frequency for the applied load. To use this feature select the second page of options for the RF channel you are using and select the Auto tune option. You will see a message popup indicating the system is tuning, this procedure can take up to 3 minutes. The system should be in open loop mode to use this procedure. The power will be se to 10% for tuning.

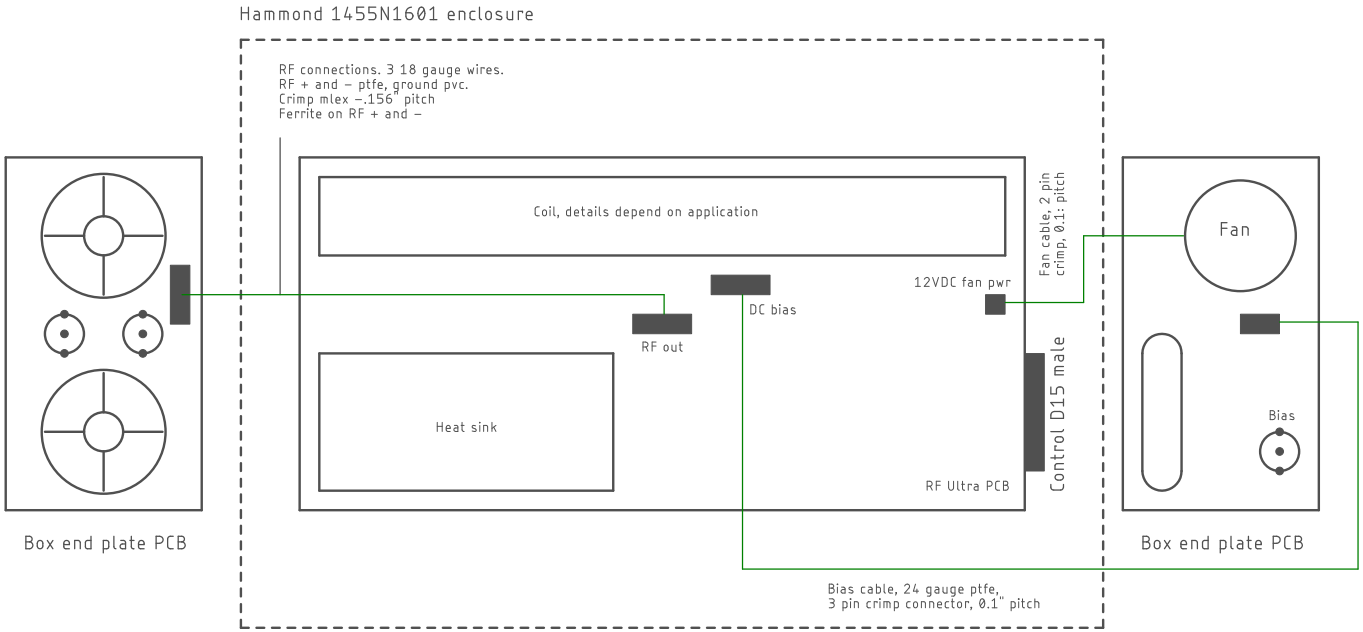
You will also find an Auto retune option. You can use this function to retune an operating system. This procedure is a bit different, the system will use the current RF drive settings and will make small frequency changes around the current operating point to reestablish the optimal operating point.

If your system requires special needs please contact us, we can build custom inductances to match your specific needs.

A number of custom configurations are possible with the RF head including the addition of a DC bias on the RF outputs and gating the RF on and off with an external signal. If you need this capability please contact us and we will help you setup your system as needed.

Warrantee

GAA Custom Electronics, LLC warrantees the RF head system to be free from defects in materials and workmanship and will repair or replace the unit for a period of one year. This warrantee assumes the system is operated in compliance with the procedures and recommendation outlines in this document. GAA Custom Electronics, LLC will also provide free phone support for up to one year. The addition of new features is not covered in this warrantee.



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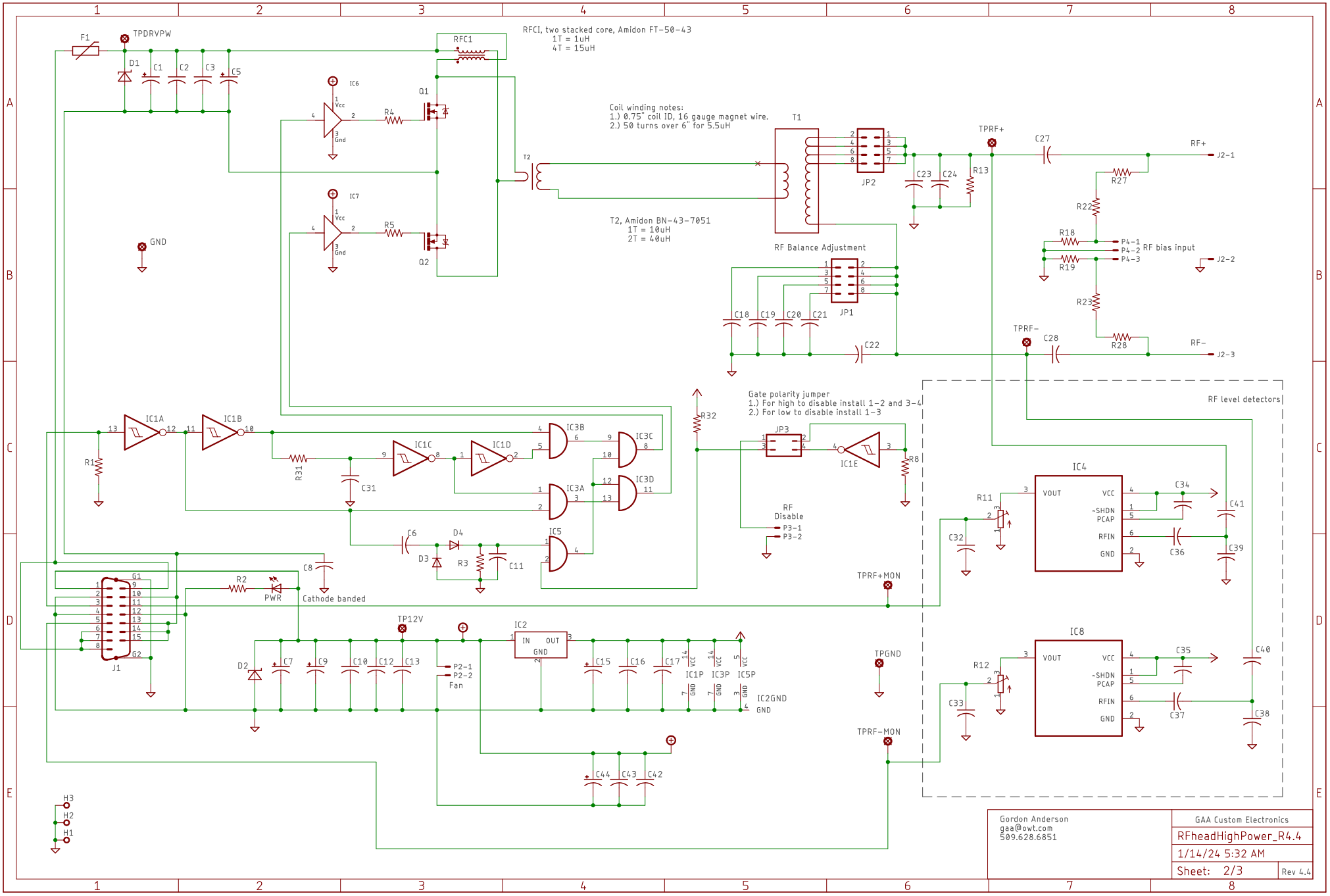
RFCI, two stacked core, Amidon FT-50-43
1T = 1uH
4T = 15uH
Coil winding notes:
1.) 0.75" coil ID, 16 gauge magnet wire.
2.) 50 turns over 6" for 5.5uH
T2, Amidon BN-43-7051
1T = 10uH
2T = 40uH
RF Balance Adjustment
RF bias input
Gate polarity jumper
1.) For high to disable install 1-2 and 3-4
2.) For low to disable install 1-3
RF level detectors
RF Disable
Cathode banded
Fan
TPDRVPW
TP12V
TPRF+
TPRF-
TPRF+MON
TPRF-MON
TPGND
RF+
RF-
IC2GND
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Coil design details:

1.) Wire diameter and turns per inch (TPI)
   - 16 gauge, 0.0513, 19 TPI
   - 18 gauge, 0.0407, 24 TPI
   - 20 gauge, 0.0323, 31 TPI
   - 22 gauge, 0.0256, 39 TPI
   - 24 gauge, 0.0203, 49 TPI
   - 26 gauge, 0.0161, 62 TPI
   - 28 gauge, 0.0127, 78 TPI
   - 32 gauge, 0.0081, 123 TPI
2.) Coil length is 5.5 inches, center taped
3.) Coil form diameter options
   - 0.25, 0.375, 0.5, 0.625, 0.75, 0.875
4.) Calculated inductance, uH

- Form diameter	0.25	0.375	0.5	0.625	0.75	0.875
- 16 gauge, 104	3.1	6.9	12.3	19.3	27.7	37.7
- 18 gauge, 132	5.0	11.2	19.9	31.0	44.7	60.8
- 20 gauge, 170	8.2	18.5	32.9	51.5	74.1	100.9
- 22 gauge, 214	13.1	29.4	52.2	81.5	117.4	159.8
- 24 gauge, 269	20.6	46.4	82.5	128.8	185.5	252.5
- 26 gauge, 341	33.1	74.5	132.5	207.0	298.1	405.8
- 28 gauge, 429	52.4	114.5	209.1	327.7	471.9	642.3
- 32 gauge, 676	130.2	292.9	520.7	813.6	1000.2	1590.0

The following coil design supports inductance taps of 10, 20, 30, and 40uH.
- Use .625" form
- First segment, 3.25" long is 16 gauge. This winding goes to the first tap point
- From the first tap to the end use 22 gauge wire.
- First tap point and high inductance side drive tap are tied together

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